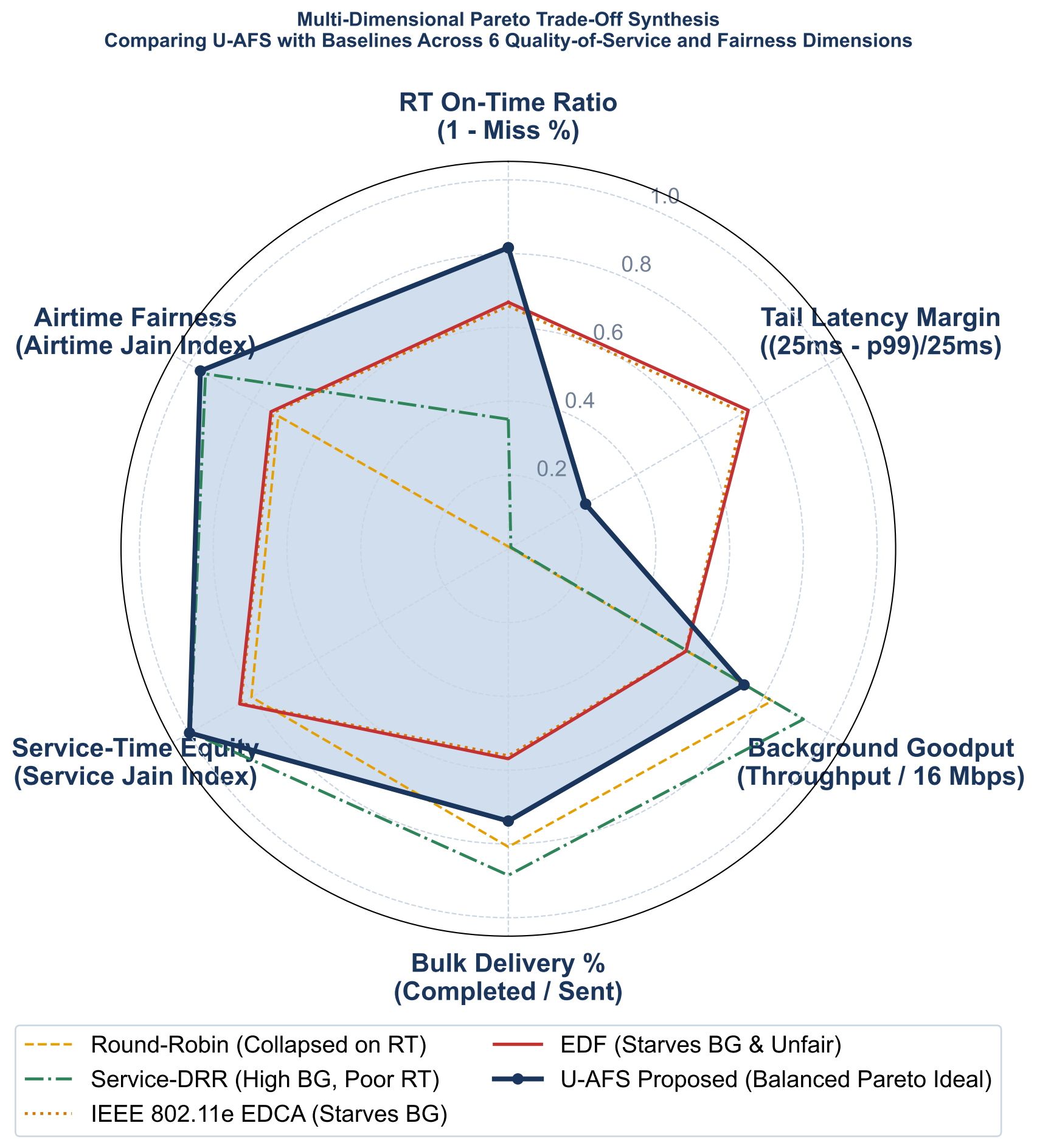
Multi-Dimensional Pareto Trade-Off Synthesis
Comparing U-AFS with Baselines Across 6 Quality-of-Service and Fairness Dimensions
RT On-Time Ratio
(1 - Miss %)
Tail Latency Margin
((25ms - p99)/25ms)
Background Goodput
(Throughput / 16 Mbps)
Bulk Delivery %
(Completed / Sent)
Service-Time Equity
(Service Jain Index)
Airtime Fairness
(Airtime Jain Index)
0.2
0.4
0.6
0.8
1.0
Round-Robin (Collapsed on RT)
Service-DRR (High BG, Poor RT)
IEEE 802.11e EDCA (Starves BG)
EDF (Starves BG & Unfair)
U-AFS Proposed (Balanced Pareto Ideal)